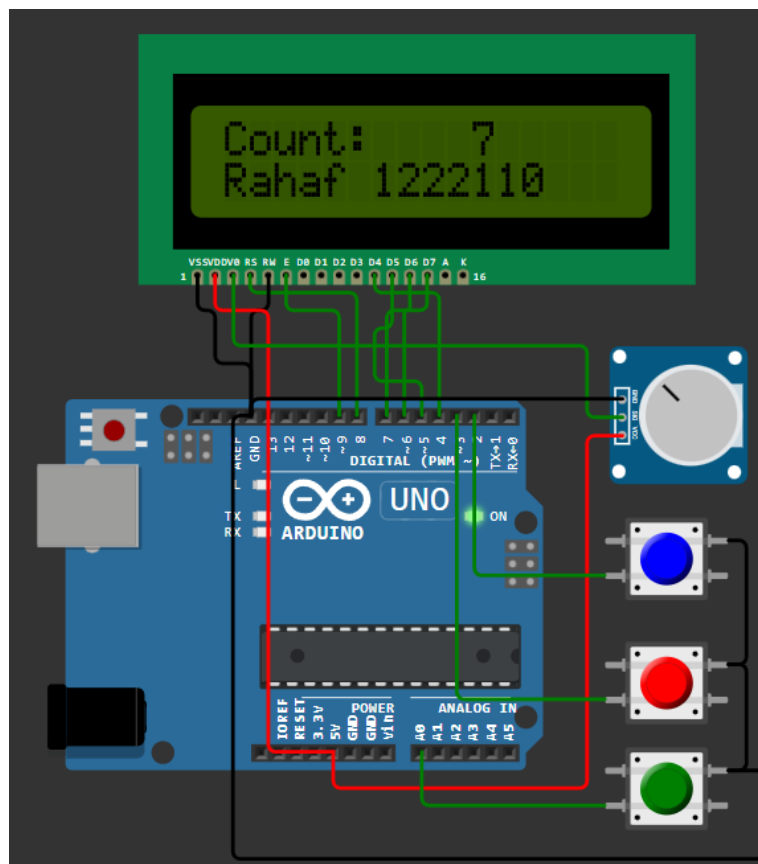


Question 4 — Hardware Interrupts: attachInterrupt() and Register-Level

Part A — Using attachInterrupt()



This project uses an Arduino Uno connected to a 16x2 LCD, a potentiometer, and three push buttons. The LCD works in 4-bit mode where RS and E are connected to pins 8 and 9, and data pins D4–D7 are connected to Arduino pins 4–7. The potentiometer is connected to the LCD V0 pin to adjust the screen contrast. Three push buttons are connected to pins D2, D3, and A0, while the other side of all buttons is connected to GND. Internal pull-up resistors are enabled in the code. The program uses the Arduino `attachInterrupt()` function for the buttons on D2 and D3. Pressing the D2 button increases the counter value, pressing the D3 button resets the counter to zero, and pressing the A0 button pauses or resumes the counter. The LCD displays the current counter value or the paused state, and the same information is also printed on the Serial Monitor.